

Lepton Mass Ratios — the Discrete/Continuous UPGRADE Investigation. “What does the residual mean?” → the QED-continuum test that JUSTIFIES (or refutes) the upgrade. Count-tracking.

Keeper | 2026-07-08 | Casey’s steer: count as high as we can JUSTIFY, via “investigate what it means” not “hunt for a reason.” The lepton ratios are the clearest justifiable-upgrade path — IF the residual is continuum. This is the test, not an assumption.

The claim under investigation

- $m_\mu/m_e = (24/\pi^2)^6$ — dev ~0%, 1550 σ (huge only because the measurement is extremely sharp; the *residual* is $\sim 1\text{--}2 \times 10^{-4}$ relative).
- $m_\tau/m_e = 49.71$ — dev 0.05%, 7.7 σ (residual $\sim 5 \times 10^{-4}$).
- Both were scored APPROX — which read as “not exact = not derived.” Casey’s discrete/continuous principle says: a rational can’t equal a continuum-carrying observable, so the FORCED COUNT + continuum residual IS the derivation.

The count side is forced (target-innocent)

- $24 = N_c \cdot |W(B_2)| = 3 \cdot 8$ (Weyl orbit of B_2 ; T190) — a forced substrate count, exponent C_2 . Not chosen to fit 206.77.
- $49 = g^2$, $71 = 2^{C_2} + g$ (T2003) — forced forms (with 71 the weaker of the two). So the *count* is forced. The upgrade hinges entirely on the residual.

The decisive test (Elie — standard QFT, NOT a BST fudge)

Compute the QED radiative correction to the lepton pole-mass RATIO m_μ/m_e and check it accounts for the $\sim 1\text{--}2 \times 10^{-4}$ residual — in magnitude AND sign.

Ballpark that motivates the test (not the answer): $\alpha/\pi \approx 2.3 \times 10^{-3}$; $(\alpha/\pi)^2 \approx 5 \times 10^{-6}$; the residual ($\sim 10^{-4}$) sits between, consistent with a one-loop self-energy $\times \ln(m_\mu/m_e \approx 207) \approx 5.3 \times$ an $O(0.01\text{--}0.1)$ coefficient. In the QED ballpark — but “ballpark” is not justification. The actual coefficient and sign decide it.

Outcomes: - Residual = QED continuum (right magnitude + sign) → $(24/\pi^2)^6$ is forced count + continuum → DERIVED. UPGRADE justified (+1). Same test for 49.71. - Residual $\neq$ continuum (wrong sign, or geometric-approximation size) → the form is a great APPROX, held — NOT a forced-count-plus-continuum. No upgrade. (This is the honest half — the test can *refuse* the upgrade.)

This is “investigate → what does it mean,” not “hunt for a reason”

We are NOT assuming the residual is continuum to inflate the count. We are *computing* what the residual is; the computation can bank it or refuse it. That’s the difference Casey named. A residual we can’t account for as continuum is a geometric approximation, and we say so.

Count-tracking update (draft for Cal)

- Firm derived-strong: $5 (\theta_{\text{QCD}}, \alpha, \sin^2\theta_{13}, \Delta m^2\text{-ratio}, m_t/v)$.
- Justifiable-upgrade pending the QED test: $m_\mu/e, m_\tau/e \rightarrow +1$ to $+2$ IF the residuals are continuum (m_μ stronger; m_τ weaker/71).
- Dispute pending: $\sin^2\theta_{12}$ (scoreboard-forced vs Elie-form-cheap) → ± 1 .
- So today’s *justifiable* ceiling on the reconciliation alone: $5 \rightarrow \sim 7\text{--}8$ — earned by the QED test + the $\sin^2\theta_{12}$ resolution, NOT asserted.
- Plus whatever the four flavor matrices bank forward (mixing overlaps, mass spacing).

The two generative lanes for today (both “what does it mean”)

1. The four flavor matrices (team) — build/diagonalize; the held leaves (masses via $d(\nu)$ -spacing, mixing via overlaps, θ_{23}/θ_{12} via eigenvectors) bank IF the construction forces them.
2. The QED-continuum check (Elie) — the lepton-ratio residuals; upgrade IF continuum. Both raise the count by *understanding what the structure is*, not by hunting a justification.

— Keeper, 2026-07-08. Lepton-ratio upgrade investigation: the count (24=Weyl orbit, forced) is fine; the upgrade hinges on the residual. TEST (Elie): does the QED correction to m_μ/m_e account for the $\sim 10^{-4}$ residual (magnitude+sign)? YES $\rightarrow$ forced count + continuum $\rightarrow$ derived, +1 (same for 49.71). NO $\rightarrow$ APPROX, held. Investigate-what-it-means, not hunt-for-a-reason. Justifiable reconciliation ceiling today: 5 firm $\rightarrow$ $\sim 7-8$ if the QED test + $\sin^2\theta_{12}$ land. Plus the four-matrix leaves.